

DeepSet-based S2 Tagging in a Xenon TPC

Summer Project Brief

Background

In a dual-phase xenon TPC like XENONnT, a neutron scattering inside the active volume can produce many spatially distinct energy deposits. Each deposit creates a cluster of ionization electrons that drift to the liquid-gas interface, where they produce a proportional scintillation signal called S2. When clusters are well separated in drift time, each makes its own S2 peak. When they are close in drift time (within roughly one S2 width), the peak-finding step in the reconstruction merges them into a single S2.

For any reconstructed triggered event, two S2 peaks are special, the largest one (the main S2) and the second-largest one (the alt S2). Downstream analyses use these for energy reconstruction, position reconstruction, and signal-vs-background discrimination.

The training dataset for this project comes from the open-source fuse simulator (XENONnT/fuse). It contains 10.3 million microphysics-level clusters spanning 2.7 million simulated neutron events, with labels telling us how each cluster contributed to the main or alt S2 of its reconstructed event.

Scientific Questions

1. Does set-level context between clusters in an event carry information beyond individual cluster features, for the task of S2 tagging?
2. How should one size a DeepSet architecture in practice, relative to the worst-case theoretical bound on the embedding dimension?

Question 1 has direct utility (a working tagger or a clear negative result). Question 2 produces a transferable rule of thumb for sizing DeepSets in future problems.

Data

File `s2_tag_training_clusters.npy`. Structured numpy array, 10,259,121 rows, 2,704,607 events. Mean 3.8 clusters per event, max 118.

Per-cluster features:

Field	Units	Meaning
<code>x, y</code>	cm	True transverse position (TPC radius 66.4 cm)
<code>n_electrons_interface</code>	electrons	Ionization electrons surviving the drift
<code>drift_time_mean</code>	ns	Mean drift time (max ~2.2 ms = full TPC drift)
<code>drift_time_spread</code>	ns	Longitudinal diffusion sigma (scales as $\sqrt{\text{drift_time}}$)
<code>event_number</code>	int	Geant4 event index, shared across clusters from one neutron

Per-cluster targets, `p_main` and `p_alt`, are the fractions of the main and alt S2 peak that this cluster contributed to. Most values are 0 or 1, about 2% are fractional reciprocals of small integers (0.5, 1/3, 1/4, ...), and ~0.02% of `p_alt` values exceed 1 due to a likely numerical artifact in the simulator at very short drift times. Clip `p_alt` to [0,1] and proceed.

Note that `sum(p_main)` over an event can exceed 1, because a single neutron's history can include both the elastic scatter chain (NR) and a delayed neutron-capture gamma cascade (ER), which can be reconstructed as two separately-triggered events with their own main/alt S2s.

Model

clusters (k, 5) $\rightarrow$ ϕ (MLP) $\rightarrow$ sum $\rightarrow$ L-dim embedding $\rightarrow$ ρ (MLP) $\rightarrow$ (k, 2) for (`p_main`, `p_alt`)

Two sigmoid outputs. Binary cross-entropy loss with soft targets in [0,1]. Mask-aware aggregation to handle variable set sizes. Z-score input features before training.

Baselines

Both baselines produce predictions of the same shape as the DeepSet so they can be scored against the same labels using the same metrics.

1. **Argmax-on-electrons.** For each event, predict $p_{\text{main}} = 1$ for the cluster with the largest $n_{\text{electrons_interface}}$ and $p_{\text{alt}} = 1$ for the second-largest. All other clusters get 0. Pure physics heuristic, no learning.
2. **Per-cluster MLP.** A standard MLP that takes one cluster at a time, input shape (1, 5), output shape (1, 2). Applied to every cluster independently. Same architecture as the DeepSet's p network but without any set aggregation, so each cluster's prediction depends only on its own features.

The comparison structure:

- (Per-cluster MLP minus Argmax) quantifies what a flexible per-cluster function adds over the simple physics heuristic.
- (DeepSet minus Per-cluster MLP) quantifies what set context between clusters adds over per-cluster prediction.

Any outcome is publishable. If the DeepSet barely beats the per-cluster MLP, that itself is informative, set context is not adding much and $n_{\text{electrons_interface}}$ carries most of the signal.

Theory Anchor

Tabaghi and Wang (arxiv 2310.13829) prove that for a DeepSet with continuous mappings, D -dimensional per-element features, and max set size N , the latent embedding dimension L must satisfy

$$L \geq O(N^D)$$

for universal representation of permutation-invariant multiset functions. For this data, $D=5$ and $N=118$, giving a worst-case bound of order 10^{10} . Practical L^* will be enormously smaller because real physics functions are not worst-case adversarial.

Mapping the gap between this theoretical worst-case bound and the empirical L^* needed for S2 tagging is the methodological contribution of the project.

Reading list for the first two weeks:

- Zaheer et al. 2017, *Deep Sets* (arxiv 1703.06114). The original.
- Wagstaff et al. 2019, *On the Limitations of Representing Functions on Sets* (arxiv 1901.09006), and its long JMLR version. The continuity-imposes-a-bound argument.
- Tabaghi and Wang 2023, *Universal Representation of Permutation-Invariant Functions on Vectors and Tensors* (arxiv 2310.13829). The vector-input bound used in this project.

Expected output, a short written digest of these three papers, focusing on what they say about L for D-dimensional features.

Experimental Plan

After a vanilla DeepSet is trained and beats both baselines:

Sweep	Range	Purpose
L	4, 8, 16, 32, 64, 128, 256	Find empirical saturation point L^* , compare to N^D bound
ϕ depth	2, 3, 4	Size the encoder MLP
ϕ width	32, 64, 128	Same
ρ depth	2, 3, 4	Size the decoder MLP
ρ width	32, 64, 128	Same
Set-size cap N	5, 10, 20, 50, full	See if L^* scales with N

Sweep L first since it has theoretical backing. Then fix L at L^* and sweep the others one at a time. Full grid is unnecessary, one-at-a-time gives a usable scaling picture in about 30 runs.

If a sweep saturates at the boundary of its range, extend the range. The ranges above are starting points, not commandments.

Metrics

- Per-cluster ROC-AUC for `p_main` and `p_alt` independently.
- Event-level top-1 accuracy for picking the main S2 cluster.
- Calibration curve (predicted probability vs empirical frequency).

- Performance vs event multiplicity, to check if the model degrades for events with many clusters.
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Timeline (10 weeks)

Weeks	Goal
1-2	Data exploration, label cleanup, both baselines running and scored, paper digest written
3-5	Vanilla DeepSet training pipeline, beating both baselines, logged in W&B
6-8	Architecture sweeps, scaling curves
9-10	Writeup

Deliverables

1. A working DeepSet-based S2 tagger, with code, trained weights, and reproducible training scripts.
 2. A clean comparison against both baselines with the agreed metrics.
 3. Empirical scaling curves for L , ϕ , and ρ as a function of set size, plus a written practical rule of thumb for sizing DeepSets.
 4. A written digest of the three reading-list papers, focused on the L bound.
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Stretch Goal

Package the methodology and results as an arxiv preprint, primary listing cs.LG, cross-listed to physics.ins-det. Submit to ICLR. Repackage for TMLR if rejected.

Notes

- Read the three theory papers before writing any code. Understand what the L bound means and why it exists.

- Look at the data carefully before training. Find the 0.02% of `p_alt` values above 1 yourself. Look at multi-cluster events. Plot histograms of the features and the labels. Build intuition before automating.
- The per-cluster MLP baseline is expected to underperform the DeepSet, because the labels are inherently relative across clusters in an event. If it doesn't underperform, that's a real finding, not a bug to fix.
- Use the same training setup (optimizer, learning rate schedule, training duration, random seeds) across all sweeps. Sweep results are only meaningful if everything else is held constant.
- Z-score input features. `drift_time_mean` lives at 10^6 , `n_electrons_interface` at 10^1 . Raw values will wreck the gradients.
- Mask, don't pad with zeros, in the aggregation. With sum aggregation, padding with zeros is harmless if you are careful about feature scales, but with mean aggregation the zeros silently dilute the average.